INCH-POUND
MIL-N-5877D
AMENDMENT 1
28 Feb 1990

MILITARY SPECIFICATION

NOZZLE, PRESSURE FUEL SERVICING, LOCKING, TYPE D-1 AND D-2, NOMINAL 2 1/2 INCH DIAMETER

This amendment forms a part of Military Specification MIL-N-5877D, dated 16 October 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

DD Form 1426 note, line 4, delete "(93)" and substitute "(53)".

PAGE 3

2.1.1, STANDARDS, MILITARY, delete "MIL-STD-143" and substitute "MIL-STD-970".

PAGE 4

3.3, Selection of specifications and standards, line 3, delete "MIL-STD-143" and substitute "MIL-STD-970".

PAGE 5

3.4, Design, line 4, delete "800 gallons" and substitute "600 gallons".

3.4.6, delete entire paragraph and substitute:

"3.4.6 Pressure drop. Sufficient line pressure shall be maintained to sustain stabilized flow in the fueling and defueling modes when tested with the fuel flowing through the nozzle and MS24484 adapter, with the MS26551 strainer removed for both modes.

3.4.6.1 Fueling mode. The pressure drop shall be not greater than 15 pounds per square inch (psi) corrected, when subjected to a 600 gpm flow of fuel and when tested in accordance with 4.6.2.2.1.

3.4.6.2 Defueling mode. With reversed fuel flow through the nozzle, the pressure drop shall be not greater than 3 psi corrected, when subjected to a 200 gpm flow of fuel and when tested in accordance with 4.6.2.2.2."

PAGE 11

4.6.2.1, Functional, line 4, delete "800 gpm" and substitute "600 gpm".

4.6.2.2, Pressure drop, delete entire paragraph and substitute:

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"4.6.2.2 Pressure drop tests. Test results shall reflect both actual and corrected pressure drops and compliance with the fueling and defueling pressure drop requirements (see 3.4.6 through 3.4.6.2). The formula for correction is as follows:

$$\Delta P_{NA} = \Delta P \times \left(\frac{1.34}{\mu}\right)_{cs}^{.25} \times \left(\frac{.769}{\rho}\right)$$

where μ = kinematic viscosity of test fluid

ρ = specific gravity of test fluid

4.6.2.2.1 Fueling mode. Using a test setup similar to Figures 1 and 4, with an inlet pressure of 40 psi, a pressure drop test shall be performed with fuel flows from 0 to 600 gpm, in 100 gpm increments.

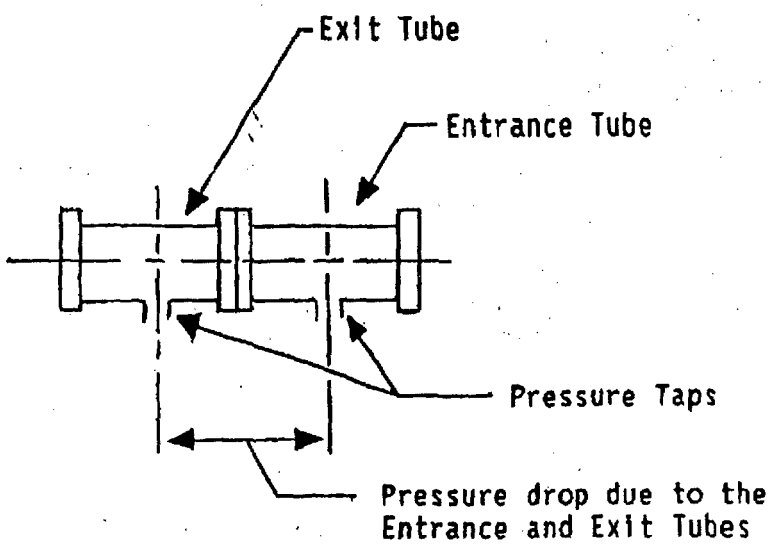
4.6.2.2.2 Defueling mode. Using the test setup specified in 4.6.2.2.1, a pressure drop test shall be performed with fuel flows from 0 to 200 gpm, in 50 gpm increments."

PAGE 14

6.3, Qualification, line 10, delete "Code 9311" and substitute "Code 5311".

PAGE 18

FIGURE 1, delete bottom drawing and substitute:



P_{NA} = Total pressure drop - pressure drop due to piping

Nominal tube size - Entrance and Exit Tube - 2 1/2" Schedule 40

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